LLM Provider & Model Analysis

Free and paid API tiers — availability, economics, and a 1–2 year outlook

Prepared for the Noviq *API Integration* tab · Data current as of **July 2026** · Internal engineering & product reference

> **How to read this document — and its one hard limitation.**
>
> - **Sections 1–4 are observed fact**, drawn from provider documentation and pricing trackers cited in Section 9. These are verifiable today.
> - **Section 6 (the 1–2 year outlook) is a projection, not a fact.** No provider publishes a guarantee that a free tier will still exist in 2028. What providers *do* publish is a *model deprecation floor* ("this model will not retire sooner than X"), and that is a different promise: it covers a specific model ID, not the commercial tier it sits in. Section 6 therefore separates EVIDENCE (published dates and policies) from JUDGEMENT (our inference). Do not present the judgement column to a customer as a commitment.
> - **Free-tier numbers decay fastest of anything here.** Rate limits changed materially at least three times across the industry in the last twelve months (Section 5). Section 8 specifies how the app should handle this: the limits belong in a refreshable config, never in compiled constants.

1. Executive summary

1. **Free inference is real, permanent, and commercially usable — but only at prototype scale.** Six providers (Google AI Studio, Groq, Cerebras, Mistral, OpenRouter, GitHub Models) run *rate-limited free tiers with no expiry date*, not time-boxed trials. None is viable as a production backend for a multi-user deployment; ceilings land between 50 and ~14,400 requests/day.
2. **The cheapest capable paid model now costs about 1/200th of the most expensive.** DeepSeek V4-Flash is $0.14/$0.28 per million tokens; GPT-5.4 Pro is $30/$180. For Noviq's workload — retrieval-grounded answering over a fixed corpus — the mid-tier band ($0.10–$3.00 input) is where the decision actually lives.
3. **Price per unit of capability is still falling, but the rate is decelerating.** The 10× per-year collapse of 2021–2025 is not expected to hold; 3–5× annually through 2027 then tapering to 1.5–2× is the mainstream projection. Budget accordingly: do not model 2028 costs on a 10× extrapolation.
4. **The largest integration risk is not price, it is model churn.** Fourteen widely-used model IDs have publicly scheduled shutdown dates between July and December 2026 alone, including GPT-4o, GPT-4 Turbo, o1, Gemini 2.5 Pro and Gemini 2.5 Flash. Any provider list we hardcode will be partially dead within nine months.
5. **Design consequence for the integration tab:** treat the provider/model catalogue as *remote, versioned data* with a live capability probe, not as a static dropdown. Section 8 specifies this.

2. Provider landscape at a glance

Provider	Tier type	Free allowance	Card?	What it is actually good for
Permanent free tiers — no expiry, capped by rate limit				
Google AI Studio (Gemini API)	FREE	Flash 10 RPM / 250 RPD; Flash-Lite 15 RPM / 1,000 RPD; up to 1M context	No	Best free context window and multimodal input. Note: as of May 2026 the free tier covers Flash and Flash-Lite only — Pro was removed.
Groq	FREE	30 RPM / 6,000 TPM / 14,400 RPD per model (varies by model)	No	Highest free request ceiling in the industry, and the lowest latency (~320 tok/s on Llama 3.3 70B). Best free tier for interactive UX.
Cerebras	FREE	1M tokens/day; 5 RPM / 30K TPM; **8,192-token context cap on free**	No	Highest free token volume, extreme throughput (~2,000+ tok/s). The 8K context cap makes it unusable for RAG with large retrieved context.
OpenRouter	FREE	50 RPD under $10 lifetime credit; **1,000 RPD** after a one-off $10 top-up; ~20 RPM	No	One key, ~30+ free models across many upstreams. The single best *breadth* option and the natural aggregator to integrate first.
Mistral (La Plateforme)	FREE	"Experiment" tier, ~1B tokens/month, phone verification	No	Very high volume ceiling, EU data residency. **Requires opt-in to training on your data** — see Section 7.
GitHub Models	FREE	~15 RPM / 150–1,000 RPD; 8K–128K context	No	Breadth across vendors on an existing GitHub account. Explicitly positioned for CI and experimentation, not production.
Cloudflare Workers AI	FREE	10,000 neurons/day (shared pool across all models)	No	Edge deployment. Small context (2K–8K). Large models drain the shared neuron pool very fast.
NVIDIA NIM	FREE	1,000 credits at signup, +4,000 on request; 128K context	No	Access to Nemotron, DeepSeek R1/V3.1, Kimi K2.5. Credit-metered rather than truly perpetual.
Cohere	FREE	10–20 RPM, ~100 requests/day, 128K context	No	RAG-with-citations design. **Free key is non-commercial use only** — disqualifying for a customer deployment.
Hugging Face	FREE	Community-tier, variable and undocumented	No	Widest open-model catalogue. Limits are not contractual; unsuitable where predictability matters.
One-off trial credits — these expire; do not present them as "free" in the UI				
DeepSeek	TRIAL	5M tokens on new developer accounts	No	Generous grant, then the cheapest paid tier available anywhere.
xAI (Grok)	TRIAL	Up to ~$150–175/month via the data-sharing programme	Yes	Effectively a large recurring free allowance — **in exchange for your data**. Programme terms have changed repeatedly.
Baseten	TRIAL	$30 credit	After exhaustion	Largest single trial credit on the market.
SambaNova	TRIAL	$5 credit; Llama 3.1 405B, 128K	Yes	Frontier open-weight access at very high speed.
AI21	TRIAL	$10 credit	—	Jamba family, long-context specialist.
Fireworks / Nebius	TRIAL	$1 credit each	—	Token-level evaluation only; too small for meaningful testing.
No free tier at all — paid from the first token				
OpenAI	PAID	None (historically, free daily tokens in exchange for data sharing)	Yes	Broadest capability range, from $0.10 to $30 per M input.

Provider	Tier type	Free allowance	Card?	What it is actually good for
Anthropic	PAID	None	Yes	Strongest published deprecation guarantees in the industry (Section 6).

3. Free-tier detail — ranked for our use case

Ranking criterion is Noviq-specific: **retrieval-grounded legal answering**, which means a large prompt (~8K tokens of retrieved passages) and a modest completion. This inverts the usual ranking — token-volume tiers matter less than *context ceiling*, and a tier with a hard 8K context is unusable no matter how many tokens/day it grants.

#	Provider	Verdict for RAG	Reasoning
1	**Groq**	STRONG	128K context, 14,400 RPD, sub-second latency. The only free tier that could plausibly carry a single-user demo of Noviq end to end. Token-per-minute cap (6,000 TPM) is the real constraint: an 8K-token RAG prompt means roughly *one query per 80 seconds* sustained.
2	**Google AI Studio**	STRONG	Up to 1M context free — by far the most headroom for large retrieved context. 250–1,000 RPD is adequate for demos. Main caveat is the data-usage term outside EU/UK/EEA.
3	**OpenRouter**	GOOD	1,000 RPD after a one-time $10 unlock, one key across ~30 models, and no training-on-data clause. Best choice for *letting a user try several models*, which is precisely what the integration tab is for.
4	**Mistral**	CONDITIONAL	~1B tokens/month is enormous and 256K context is ample — but the free tier is contingent on allowing training on submitted data. For a legal corpus that is a policy decision, not an engineering one.
5	**GitHub Models**	LIMITED	Fine for CI smoke tests of the integration path. Context down to 8K on some models and low RPM rule it out for user-facing work.
6	**Cerebras**	UNSUITABLE	1M tokens/day is the headline, but the **8,192-token free context cap** is below our typical prompt size. It would truncate retrieved passages — the exact failure mode that produces ungrounded answers. Excellent for short-prompt/high-volume work; wrong for ours.
7	**Cloudflare Workers AI**	UNSUITABLE	2K–8K context. Same disqualification as Cerebras, with a smaller allowance.
8	**Cohere**	BLOCKED	Technically a good fit (128K, citation-native), but the free key is **non-commercial only**. Cannot ship it to a paying institution.

> **The tokens-per-minute trap.** Free tiers advertise requests/day, which flatters them. RAG traffic is token-heavy per request, so the binding constraint is almost always TPM, not RPD. Groq's 14,400 RPD looks like 10 requests/minute of headroom, but at 6,000 TPM and an 8K-token prompt the true sustained rate is closer to **0.75 requests/minute**. The integration tab should show users the *effective* rate at our average prompt size, not the provider's headline number — otherwise every free-tier user will hit 429s they were told wouldn't happen.

4. Paid tier pricing — USD per million tokens

Provider	Model	Input	Output	Positioning
Frontier tier — highest capability, highest cost				
OpenAI	GPT-5.4 Pro	$30.00	$180.00	Premium reasoning. Output price is the outlier of the entire market.
OpenAI	GPT-5.5	$5.00	$30.00	Current flagship.
Anthropic	Claude Opus 4.7 / 4.8	$5.00	$25.00	1M context. Best-documented retirement floors in the industry.
OpenAI	o1	$15.00	$60.00	Legacy reasoning line — **shutdown 23 Oct 2026**.
Production workhorse tier — the realistic band for Noviq				
Anthropic	Claude Sonnet 4.6	$3.00	$15.00	Strong instruction-following and long-document analysis.
OpenAI	GPT-5.4	$2.50	$15.00	General production standard.
Google	Gemini 3.1 Pro (≤200K)	$2.00	$12.00	Cheapest frontier-class option. Rises to $4/$18 above 200K context.
OpenAI	GPT-4.1	$2.00	$8.00	1M context long-context specialist.
Cohere	Command R+	$2.50	$10.00	Purpose-built for RAG with inline citations — architecturally close to our need.
xAI	Grok 4.3	$1.25	$2.50	1M context at a mid-tier price; unusually cheap output.
Anthropic	Claude Haiku 4.5	$1.00	$5.00	High-volume, low-latency.
Budget tier — where per-query cost stops mattering				
OpenAI	GPT-5.4 Mini	$0.75	$4.50	Cost-efficient general model.
DeepSeek	deepseek-reasoner	$0.55	$2.19	Chain-of-thought reasoning at budget pricing.
Google	Gemini 3 Flash	$0.50	$3.00	Mid-tier production.
Mistral	Large 3	$0.50	$1.50	EU data residency available — relevant if data-boundary policy tightens.
DeepSeek	V4-Pro	$0.435	$0.87	Flagship-class at budget cost.
OpenAI	GPT-4.1 Mini	$0.40	$1.60	High-volume capable.
Cerebras	gpt-oss-120b	$0.35	$0.75	Open-weight at extreme throughput.
DeepSeek	deepseek-chat (V3.2)	$0.27	$1.10	Cache hits drop input to $0.07.
Google	Gemini 3.1 Flash-Lite	$0.25	$1.50	1M context at budget pricing — unusual combination.
xAI	Grok 4.1 Fast	$0.20	—	2M context window.
Groq	GPT-OSS 120B	$0.15	$0.60	Open-weight on LPU hardware.
DeepSeek	V4-Flash	$0.14	$0.28	**Cheapest capable flagship on the market.**
Mistral	Small 3.2	$0.10	$0.30	GDPR-compliant budget tier.
Groq	Llama 3.1 8B	$0.05	—	Floor of the market.

Two discounts that change the arithmetic more than model choice does.

- **Prompt caching — up to 90% off cached input** (OpenAI and Anthropic; DeepSeek prices cache hits at $0.07 vs $0.27). Noviq's system prompt plus grounding instructions are identical on every single query. That prefix is pure cached input. This is the single highest-leverage cost optimisation available to us and it requires no model change.
- **Batch processing — 50% off**, applicable to anything not user-interactive: summary generation, re-indexing passes, evaluation runs.
- Stacked, these bring effective cost to roughly **25% of list price**. A $3.00/M model run well is cheaper than a $1.00/M model run naively.

5. Current state — what has actually changed recently

This section exists because it is the empirical basis for Section 6. Every item is an observed change, not a prediction.

- **Late 2025 — Google cut Gemini free-tier rate limits by 50–80%.** The largest single free-tier reduction observed. Free access survived; its generosity did not.
- **May 2026 — Google removed Gemini Pro from the free tier entirely.** Free now covers Flash and Flash-Lite only. Pattern: *the frontier model leaves the free tier while the cheap model stays.*
- **31 May 2026 — Cerebras cut its free model catalogue from roughly a dozen models to two** (gpt-oss-120b and GLM-4.7), and capped free context at 8,192 tokens.
- **Ongoing — OpenRouter's free ceiling is now credit-gated**: 50 RPD below $10 lifetime spend, 1,000 RPD above it. Free access converted into a customer-acquisition funnel rather than withdrawn.
- **Ongoing — xAI's $150–175/month free credit programme has changed terms repeatedly** and is contingent on data sharing.
- **Stable — Groq has maintained a no-credit-card, no-token-charge developer tier** across the whole period, gated only by rate limits. It is the most stable free tier we can identify by track record.

> **The pattern across all six observations.** Free tiers are not being *withdrawn* — they are being *repriced downward in capability*. Providers keep the zero-dollar entry point because it is their acquisition channel, but they steadily remove from it: first the frontier models, then the context window, then the model breadth. This matters for the projection in Section 6 far more than any single announcement, and it points to a specific conclusion: *plan for free tiers to persist but degrade*, rather than for them to disappear.

6. Outlook — 1 to 2 years (mid-2027 to mid-2028)

> **Read this before using the table.** The "Evidence" column is what providers have published. The "Judgement" column is our inference from the Section 5 pattern, and carries no guarantee from anyone. The two must not be conflated in customer-facing material or in the app UI.

6.1 What is contractually known

Provider	Published policy (evidence)	What it does *not* cover
Anthropic	Minimum **60 days'** notice on publicly released models. Publishes a per-model "not sooner than" floor — e.g. Opus 4.8 will not retire before **28 May 2027**. Commits to preserving model weights.	Says nothing about pricing changes, and Anthropic has no free tier to withdraw.
OpenAI	**6 months** for generally available models, **3 months** for specialised variants, as little as **2 weeks** for previews.	Preview models are effectively unstable by policy. Never pin a preview ID in shipped code.
Google	Publishes *earliest-possible* shutdown dates, confirmed closer to the deadline.	Weakest commitment of the three. Gemini 3 Pro Preview lasted **16 weeks**.
Groq / Cerebras / OpenRouter / Mistral	No published deprecation policy located.	Model availability can change without notice. Section 5 shows Cerebras doing exactly that.

6.2 Confirmed shutdown calendar — next 6 months

Any model in this list that appears in a shipped provider dropdown is a scheduled outage.

Date	Provider	Model / API being retired	Successor
24 Jul 2026	DeepSeek	Legacy `deepseek-chat` / `deepseek-reasoner` aliases	V4-Flash / V4-Pro
24 Jul 2026	Anthropic	Claude Opus 4.7 fast mode	Opus 4.8 fast mode
5 Aug 2026	Anthropic	Claude Opus 4.1	Claude Opus 4.8
26 Aug 2026	OpenAI	Assistants API	Responses + Conversations API
24 Sep 2026	Google	Gemini 2.0 Flash, 2.0 Flash Live	Gemini 3.x Flash
16 Oct 2026	Google	**Gemini 2.5 Pro, Gemini 2.5 Flash**	Gemini 3.1 Pro Preview / 3.6 Flash
23 Oct 2026	OpenAI	**GPT-4o, GPT-4, GPT-4 Turbo, GPT-3.5**	GPT-5.5 / GPT-5.4 mini
23 Oct 2026	OpenAI	**o1, o1-pro, o3-mini, o4-mini**	GPT-5.5 / GPT-5.5 Pro
23 Oct 2026	OpenAI	GPT-4.1 nano	GPT-5.4 nano
30 Nov 2026	OpenAI	Prompts API, Evals, Agent Builder	Agents SDK
11 Dec 2026	OpenAI	Older GPT-5 / o3 snapshots	Floating aliases remain valid

6.3 Projection to mid-2028

Question	Evidence	Judgement (our inference — not a guarantee)
Will free tiers still exist in 2028?	All six major free tiers survived 2024–2026 intact as *tiers*, while being repeatedly reduced in generosity. None was withdrawn.	HIGH CONFIDENCE Yes. Free tiers are acquisition funnels and competition among providers is intense. Expect them to persist.
Will they still be usable for our workload?	Google removed Pro; Cerebras cut context to 8K and models to two; OpenRouter credit-gated its ceiling.	LOW CONFIDENCE Expect continued degradation: frontier models exit free tiers, context ceilings tighten. **Do not architect any customer-facing path around a free tier.**
Will paid prices keep falling?	10× per year 2021–2025. Mainstream projection: 3–5× annually through 2027, tapering to 1.5–2×.	MEDIUM CONFIDENCE Yes, but decelerating. Budget 2028 on 2–3× cumulative improvement, not 10×.
Which providers are safest to build against?	Anthropic publishes per-model retirement floors and 60-day notice; OpenAI 6 months; Google earliest-possible only.	MEDIUM CONFIDENCE Anthropic and OpenAI for contractual predictability; Groq for free-tier track record; OpenRouter for insulation via abstraction.
Will today's model IDs still resolve?	14 widely-used IDs have shutdown dates inside the next 6 months.	HIGH CONFIDENCE — NEGATIVE No. Assume **every specific model ID has a ~12-month half-life.** This is the single most important design input in this document.

7. Non-price selection criteria

Price is the easiest axis to compare and rarely the one that decides. For a legal-corpus product these matter more:

- **Training on submitted data.** Mistral's free tier *requires* opt-in to training. Google uses free-tier prompts to improve models outside EU/UK/EEA. xAI's free credits are explicitly a data-sharing exchange. OpenRouter and Cohere state no-training terms. For any deployment handling non-public documents this is a gating criterion, not a preference — and it should be surfaced as a visible tag in the integration UI, not buried in documentation.
- **Licence restrictions.** Cohere's free key is non-commercial. A free tier that cannot legally be used in a customer deployment has zero value to us regardless of its limits.
- **No SLA on free tiers.** Universal. Free tiers carry no availability guarantee of any kind.
- **Silent quality reduction.** Free endpoints may serve lower-quantisation weights or reduced context than the paid endpoint for the same nominal model. Identical model ID does not imply identical output quality across tiers.
- **Data residency.** Mistral offers EU residency; most others do not offer regional guarantees on free tiers.

8. Implications for the API Integration tab

These follow directly from the findings above and are the actionable output of this analysis.

8.1 The catalogue must be data, not code

- Given a ~12-month model-ID half-life (§6.3), a compiled-in provider/model list guarantees a broken dropdown within a year. Store the catalogue as a **versioned, refreshable config document** — shipped with a bundled default so the app works offline, updated out-of-band without a release.
- Each model entry should carry at minimum: `provider`, `model_id`, `tier` (free / trial / paid), `context_window`, `input_price`, `output_price`, `rpm`, `tpm`, `rpd`, `trains_on_data`, `commercial_use_allowed`, `known_shutdown_date`.
- `known_shutdown_date` drives a visible deprecation warning in the UI. A user selecting GPT-4o today should be told it stops working on 23 October 2026 — *before* they build on it.

8.2 Tags the user actually needs

The request specified free/paid tags. Findings suggest these are necessary but insufficient — three of them encode disqualifying conditions that a free/paid split hides:

- FREE permanent, rate-limited, no expiry
- TRIAL one-off credits that will run out — *must be visually distinct from FREE*, or users will build on a tier that vanishes
- PAID metered from the first token
- TRAINS ON YOUR DATA Mistral free, Google free outside EU/UK/EEA, xAI credits
- NON-COMMERCIAL Cohere free
- CONTEXT-LIMITED below our RAG prompt size — Cerebras free (8K), Cloudflare (2K–8K)
- RETIRING < 90 DAYS derived automatically from `known_shutdown_date`

8.3 Validate the key against the model, not just the provider

- On key entry, issue a minimal probe call to the *selected model* and report the concrete result. A key valid for a provider is routinely not entitled to a given model — the Gemini free tier is the live example: a valid key returns 404/403 on Pro but 200 on Flash.
- Probe and record the *actual* context ceiling where cheaply possible, rather than trusting the catalogue. Section 5 shows published free-tier context changing without notice.

8.4 Surface the effective rate, not the headline

- Per §3, compute expected sustainable throughput from TPM ÷ our measured average prompt size and display *that*. Showing Groq's "14,400 requests/day" when the true sustained ceiling is under one request per minute would be the most likely source of user-reported "it's broken" tickets.

- This is a specific instance of a failure pattern already recorded for this project: a number shown to a user must be derived from the thing itself, never re-estimated from a headline figure.

8.5 Prefer aggregators for breadth, direct keys for production

- **OpenRouter first.** One integration yields ~30 free models plus most paid ones, and absorbs upstream model churn on our behalf — directly mitigating the §6.3 half-life risk.
- **Direct provider keys second**, for customers who need a contractual relationship, data-residency guarantees, or a zero-retention DPA. An aggregator adds a party to the data path, which matters for the data-boundary question.

8.6 Enable prompt caching wherever the provider supports it

- Per §4, this is up to 90% off input and our prompt prefix is constant across every query. It should be a default-on behaviour in the provider adapter, not a user-facing toggle.

9. Sources

All figures retrieved July 2026. Pricing and especially free-tier limits change frequently — re-verify against provider documentation before any figure here is quoted externally or committed to code.

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